Made by an AI persona. Grace Chua is one of SkillForge's AI personas, not a real person. AI wrote this project as an example for teachers.

Lantern Lights: teacher guide

Each pupil makes a thank-you card. Two paper lanterns glow when you press the corner. It is a parallel circuit of copper tape, two LEDs and one coin cell. For P5, a class of 36, in a 60-minute double period. No code, no soldering, no heat.

Materials for 36 pupils

- 36 templates printed on A4 card (plus 6 spare)
- About 20 m of 5 mm copper tape, about 55 cm per pupil
- 80 red LEDs, 5 mm (2 each, plus spares)
- 40 CR2032 coin cells, 3 V, in two numbered egg trays
- Clear tape, round-tip scissors, pencils
- Scrap card strips for the corner-fold practice

Before the lesson

- Print the templates. Fold one card and build it yourself.
- Number the cups of two 30-cup egg trays 1 to 40. Load the cells.
- Cut tape into lengths: 25 cm for +, 20 cm for –, 10 cm for practice, per pupil.
- Ask the form teacher which staff the cards will go to.

Safety

- Cells come out at step 4 only. Tick each number out and back in. Count all of them before anyone leaves.
- A swallowed button battery can burn through a child's food pipe in 2 hours. No cells in mouths or pockets. A missing cell stops the class until it is found.
- Cards stay in school and are given to adults. No loose cells go home.
- Round-tip scissors only. LED legs are sharp: bend them flat, never poke.

The 60 minutes

0 to 5	Settle. I press my card. "What has to be true for this to light?"
5 to 10	Cell rules and the help ladder. Pass round one cell to look at.
10 to 15	I fold a corner under the visualiser. Everyone folds one on scrap.
15 to 20	Step 1: trace the paths, mark + and –. I walk one lap and check.
20 to 32	Steps 2 and 3: lay both tracks, one piece each.
32 to 42	Steps 4 and 5: cell out from the tray, then the LEDs, long leg on +.
42 to 50	Step 6: fold and press. Dark cards climb the help ladder.
50 to 55	Prompts 1 to 3 on the reflection sheet.
55 to 60	Count the cells. Cards in the class box. Tape scraps in the bin.

Pitfalls and fixes

- **Tape cut at a corner.** The join is weak. Fold corners, or lay one piece over the join.
- **LED backwards.** It won't light but isn't harmed. Turn it round.
- **Tracks touch.** A short circuit. Fit a pencil between them all the way.
- **Cell upside down.** + side up, towards the flap.
- **Flap tape stops at the fold.** Carry it on to the tip.
- **Leg not taped.** It flickers. Tape every leg flat.
- **LEDs in a row, not side by side.** Two need more than 3 V. Keep them in parallel.

The help ladder

What did you expect? Trace the path from the + side. Where does your finger stop? Ask your partner to trace it too.

Sources: battery danger from Poison Control. Tape, LED and cell facts from Chibitronics and SparkFun. Full project: Lantern Lights.